



International  
Institute of Information  
Technology Bangalore

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# **COM 801 Project Elective**

## **PROJECT REPORT**

### **Data Analysis and RAG**

### **Recommendation System**

### **For Compost Monitoring System**

**Instructor:**

**Prof. Jyotsna Bapat**

**and**

**Prof. Sasirekha GVK**

**By:**

**C Siva Kumar**

**MT2024514**

# 1. Introduction

## Objective

Our project focuses on analyzing sensor data from two composting cycles to understand the composting process and build a Retrieval-Augmented Generation (RAG) system for the Compost Monitoring System. The goal is to provide actionable recommendations for optimizing composting efficiency, reducing harmful emissions ( $\text{NH}_3$ ,  $\text{H}_2\text{S}$ ,  $\text{CH}_4$ ), and improving compost quality.

## Scope

- **Data Analysis:**
  - Collected and analyzed sensor data from two composting cycles:
    - **Cycle 1:** Baseline composting process (40+ days).
    - **Cycle 2:** Improved composting setup with enhanced aeration (40+ days).
- **RAG System:** Developed a prototype system to retrieve contextually relevant composting guidelines from research papers and generate recommendations.

## Parameters Monitored

- **Gases:**  $\text{NH}_3$ ,  $\text{H}_2\text{S}$ ,  $\text{CH}_4$  (emissions control).
- **Environmental Factors:** Soil moisture, ambient temperature, humidity.
- **Compost Health:** Compost temperature, pH value.



Fig. 1 Composting Bin Setup

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## 2. Part A: Data Analysis

### 2.1 Data Processing

#### Data Collection

Sensor data was collected wirelessly using a Raspberry Pi. Daily readings were converted into Excel files via a Python script (this was done by Arun Sir) and later standardized into CSV files.

#### Preprocessing

- **Missing Values:** Applied interpolation methods:

| Parameter                       | Method                | Reasoning                              |
|---------------------------------|-----------------------|----------------------------------------|
| NH <sub>3</sub> , Soil Moisture | Linear                | Stable trends over time.               |
| H <sub>2</sub> S                | Cubic Spline          | Captured non-linear fluctuations.      |
| CH <sub>4</sub> , pH            | Polynomial (Order 2)  | Modelled gradual changes in emissions. |
| Humidity                        | Linear + Rolling Mean | Smoothed short-term noise.             |

- **Outlier Removal:** Used the Interquartile Range (IQR) method. It is also known as Box-Whisker plot.

### 2.2 Data Aggregation

Aggregated daily sensor readings into two datasets:

- **Cycle 1:** Baseline composting process.
- **Cycle 2:** Improved setup with better aeration.  
Calculated daily averages for each parameter to reduce noise.

## 2.3 Data Visualization and Insights

Tools: Pandas, Matplotlib, NumPy.

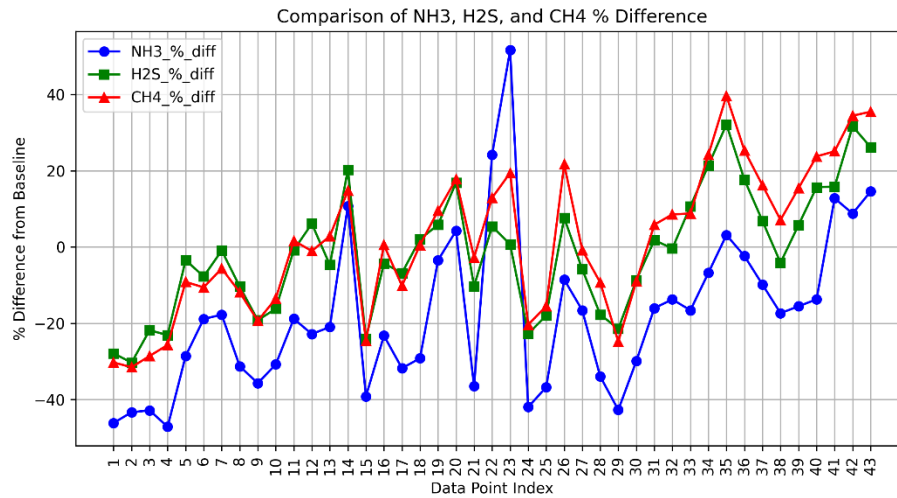


Fig 2. Cycle 1

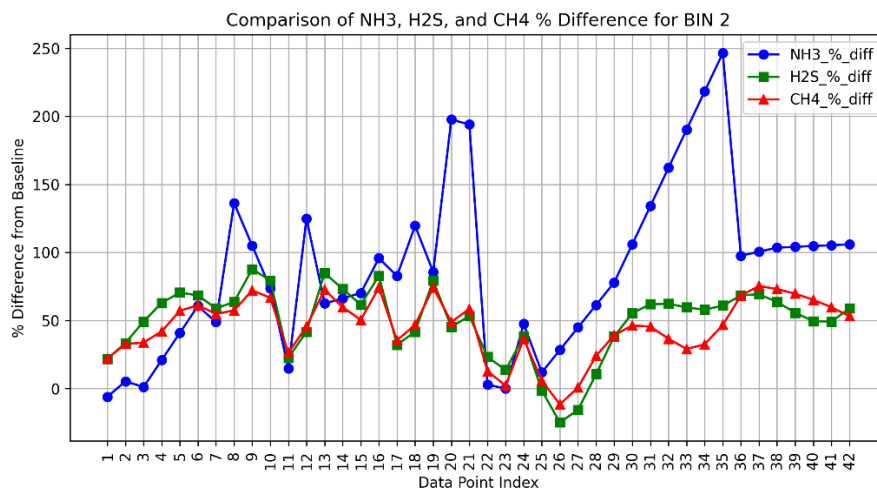


Fig 3. Cycle 2

### Key Findings

- **Cycle 1:**
  - NH<sub>3</sub> and CH<sub>4</sub> correlation: 0.83 (indicating anaerobic conditions).
  - Stalled microbial activity.
  - **Insight:** Poor aeration slowed decomposition.
- **Cycle 2:**
  - Compost temperature stabilized at ~30°C by Day 30 (25% faster than Cycle 1).
  - Reduced CH<sub>4</sub> and H<sub>2</sub>S spikes.

- **Insight:** Improved aeration reduced the anaerobic conditions.
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### 3. Part B: RAG-Based Recommendation System

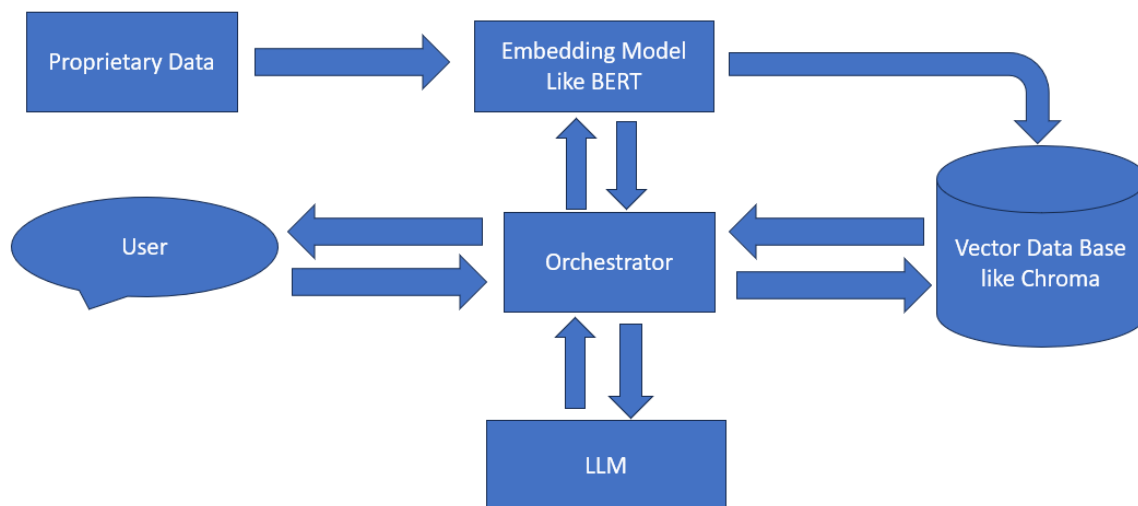


Fig 4. Level 1 diagram of the RAG System

#### 3.1 Data Sources

Scraped composting guidelines from PDFs of research papers related to compost monitoring systems, composting practices etc. stored in Google Drive.

#### 3.2 Data Processing Pipeline

##### 1. Text Extraction

Used pdfplumber python library to preserve document structure.

##### 2. Cleaning

- Removed citations (e.g., "[3]") and URLs.
- Retained section headers using regex.

##### 3. Chunking

Split text into 1,000-character chunks with 200-character overlap using Recursive Character Text Splitter.

##### 4. Embedding

Generated embeddings with BAAI/bge-small-en-v1.5 model.

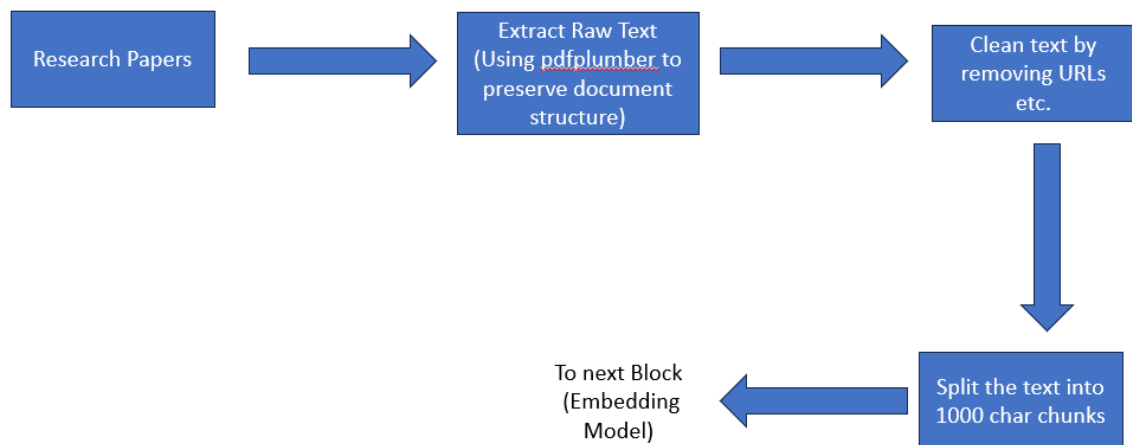


Fig 5. Preprocessing Pipeline

## 5. Vector Database

### Why Chroma?

- Chose ChromaDB for prototyping simplicity:
  - Stores embeddings + metadata (e.g., section titles).
  - Enables fast similarity searches.
  - Persists data to disk for reuse.

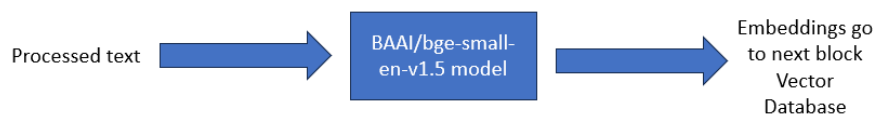


Fig. 6. Creating Embeddings

## 3.3 Current System Output

- Functional ChromaDB database with 1,200+ text chunks.
- Metadata tags for filtering (e.g., "Aeration", "pH Control").
- Sample query output for retrieving relevant composting guidelines.

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## 4. Conclusion

### Data Analysis Insights

Cycle 2 demonstrated a 25% faster composting time with improved aeration, validating its role in emission reduction.

### RAG System Progress

Built an end-to-end pipeline for knowledge retrieval.

### Next Steps:

- Integrate LLMs (e.g. GPT-4 or open source ones like Llama) for natural language recommendations.
- Link sensor data trends to real-time advice (e.g., pH adjustments).
- Scale with MongoDB/Milvus/Qdrant for production.

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### Final Notes

This project combines data-driven composting analysis with AI-powered recommendations, offering a sustainable waste management blueprint.

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